Matgeo Presentation - Problem 2.5.29

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Problem Statement

The points $\mathbf{A} = (4, 7),\ \mathbf{B} = (p, 3),\ \mathbf{C} = (7, 3)$ are the vertices of a right triangle, right-angled at $\mathbf{B}$. Find the value of p.

Solution: Condition for Right Angle at B

Let

$$\mathbf{A} = \begin{pmatrix} 4 \\ 7 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} p \\ 3 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 7 \\ 3 \end{pmatrix} \tag{0.1}$$

Right angle at $\mathbf{B} \implies$ the dot product of the adjacent sides at $\mathbf{B}$ is zero:

$$(\mathbf{A} - \mathbf{B})^T (\mathbf{C} - \mathbf{B}) = 0 \tag{0.2}$$

Solution:

Compute the vectors:

$$\mathbf{A} - \mathbf{B} = \begin{pmatrix} 4 - p \\ 4 \end{pmatrix}, \qquad \mathbf{C} - \mathbf{B} = \begin{pmatrix} 7 - p \\ 0 \end{pmatrix} \tag{0.3}$$

Their dot product:

$$(\mathbf{A} - \mathbf{B})^T (\mathbf{C} - \mathbf{B}) = \begin{pmatrix} 4 - p \\ 4 \end{pmatrix}^T \begin{pmatrix} 7 - p \\ 0 \end{pmatrix} \tag{0.4}$$

$$= (4 - p)(7 - p) + 4 \cdot 0 \tag{0.5}$$

$$= (4 - p)(7 - p) \tag{0.6}$$

Solution:

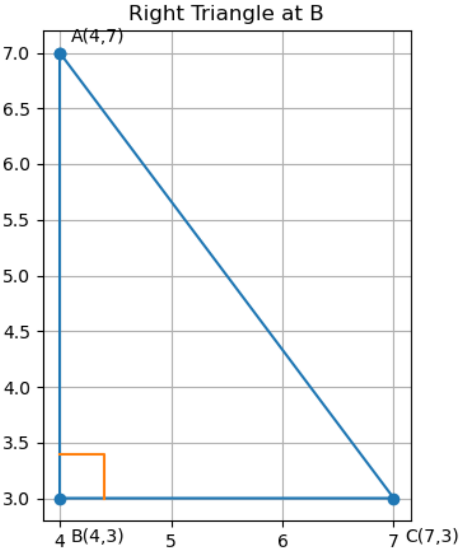
Right angle condition gives:

$$(4-p)(7-p)=0 \;\Rightarrow\; p=4 \text{ or } p=7 \tag{0.7}$$

If $p=7$, then $\mathbf{B}=\mathbf{C}$ (degenerate). Hence,

$$\boxed{p=4} \tag{0.8}$$

Result Plot



C Source Code: points.c

```c
// geom.c
#include <stdio.h>

int is_right_at_B(int Ax, int Ay, int Bx, int By, int Cx, int Cy) {
    long BAx = Ax - Bx, BAy = Ay - By;
    long BCx = Cx - Bx, BCy = Cy - By;
    long dot = BAx * BCx + BAy * BCy;
    return dot == 0 ? 1 : 0;
}

int solve_p_for_given_question(void) {
    // A(4,7), B(p,3), C(7,3); right angle at B  (4-p,4)(7-p,0)=0  p=4
    return 4;
}
```

Python Script: call c.py

```
# use_geom.py
import ctypes
from ctypes import c_int

lib = ctypes.CDLL("./libgeom.so")

lib.is_right_at_B.argtypes = [c_int, c_int, c_int, c_int, c_int, c_int]
lib.is_right_at_B.restype = c_int

lib.solve_p_for_given_question.argtypes = []
lib.solve_p_for_given_question.restype = c_int

p = lib.solve_p_for_given_question()
print(p)

ok = lib.is_right_at_B(4, 7, p, 3, 7, 3)
print("RIGHT" if ok else "NOT_RIGHT")
```

Python Script: plot.py

```
# triangle_plot.py
# Plot A(4,7), B(p,3), C(7,3) with p=4 (right angle at B)

import math
import matplotlib.pyplot as plt

# Points
A = (4, 7)
p = 4
B = (p, 3)
C = (7, 3)

# Unpack for plotting
xs = [A[0], B[0], C[0], A[0]]
ys = [A[1], B[1], C[1], A[1]]

# Draw triangle
plt.figure(figsize=(5, 5))
plt.plot(xs, ys, marker='o')

# Labels
plt.text(A[0]+0.1, A[1]+0.1, "A(4,7)")
plt.text(B[0]+0.1, B[1]-0.4, f"B({p},3)")
plt.text(C[0]+0.1, C[1]-0.4, "C(7,3)")
```

Python Script: plot.py

```
# Indicate right angle at B by plotting little square
# Vector BA and BC
BA = (A[0]-B[0], A[1]-B[1])
BC = (C[0]-B[0], C[1]-B[1])
# Small step along each to draw the corner
scale = 0.4
p1 = (B[0] + scale * (BA[0]/math.hypot(*BA)), B[1] + scale * (BA[1]/math.hypot(*BA)))
p2 = (B[0] + scale * (BC[0]/math.hypot(*BC)), B[1] + scale * (BC[1]/math.hypot(*BC)))
corner = (p1[0] + (p2[0]-B[0]), p1[1] + (p2[1]-B[1]))
plt.plot([p1[0], corner[0], p2[0]], [p1[1], corner[1], p2[1]])

plt.gca().set_aspect('equal', adjustable='box')
plt.grid(True)
plt.title("Right␣Triangle␣at␣B")

# Always save to a file so you have output even without a display
plt.savefig("triangle.png", dpi=150)
```